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Fin structures having varied fin heights for semiconductor device

Abstract

A method of forming first and second fin field effect transistors (finFETs) on a substrate includes forming first and second fin structures of the first and second finFETs, respectively, on the substrate. The first and second fin structures have respective first and second vertical dimensions that are about equal to each other. The method further includes modifying the first fin structure such that the first vertical dimension of the first fin structure is smaller than the second vertical dimension of the second fin structure and depositing a dielectric layer on the modified first fin structure and the second fin structure. The method further includes forming a polysilicon structure on the dielectric layer and selectively forming a spacer on a sidewall of the polysilicon structure.

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 16/746,547, titled “Fin Structures Having Varied Fin Heights For Semiconductor Device,” filed Jan. 17, 2020, which is a continuation of U.S. patent application Ser. No. 16/023,640, titled “Fin Structures Having Varied Fin Heights For Semiconductor Device,” filed Jun. 29, 2018, which is a divisional of U.S. patent application Ser. No. 15/724,519, titled “Fin Structure Having Varied Fin Heights for Semiconductor Device,” filed Oct. 4, 2017, which claims the benefit of U.S. Provisional Patent Application No. 62/552,236, titled “Fin Structure for Semiconductor Device,” filed Aug. 30, 2017, each of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor

devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs and fin field effect transistors (finFETs). Such scaling down has increased the complexity of semiconductor manufacturing processes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1A-1B and 2A-2B are isometric views of fin field effect transistors (finFETs), in accordance with some embodiments.

(3) FIG. 3 is flow diagram of a method for fabricating a finFET, in accordance with some embodiments.

(4) FIGS. 4A-13A are isometric views of a finFET at various stages of its fabrication process, in accordance with some embodiments.

(5) FIGS. 4B-13B are isometric views of a finFET, in accordance with some embodiments.

(6) Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used

to pattern the fins.

(10) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(11) It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

(12) As used herein, the term “selectivity” refers to the ratio of the etch rates of two materials under the same etching conditions.

(13) The term “about” as used herein indicates the value of a given quantity varies by $\pm 10\%$ of the value, unless noted otherwise.

(14) As used herein, the term “substrate” describes a material onto which subsequent material layers are added. The substrate itself may be patterned. Materials added on top of the substrate may be patterned or may remain unpatterned. Furthermore, the substrate may be a wide array of semiconductor materials such as, for example, silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate may be made from an electrically non-conductive material such as, for example, a glass or a sapphire wafer.

(15) As used herein, the term “high-k” refers to a high dielectric constant. In the field of semiconductor device structures and manufacturing processes, high-k refers to a dielectric constant that is greater than the dielectric constant of SiO₂ (e.g., greater than 3.9).

(16) As used herein, the term “low-k” refers to a small dielectric constant. In the field of semiconductor device structures and manufacturing processes, low-k refers to a dielectric constant that is less than the dielectric constant of SiO₂ (e.g., less than 3.9).

(17) As used herein, the term “p-type” defines a structure, layer, and/or region as being doped with p-type dopants, such as, for example, boron.

(18) As used herein, the term “n-type” defines a structure, layer, and/or region as being doped with n-type dopants, such as, for example, phosphorus.

(19) As used herein, the term “vertical” means nominally perpendicular to the surface of a substrate.

(20) As used herein, the term “critical dimension” refers to the smallest feature size (e.g., line width) of a finFET and/or an element of an integrated circuit.

(21) As used herein, the term “substantially” indicates that the value of a given quantity varies by $\pm 1\%$ to $\pm 5\%$ of the value.

(22) This disclosure provides example structures and methods for simultaneously fabricating semiconductor devices having different fin structures on a same substrate.

(23) FIG. 1A is an isometric view of a device **100A**, according to some embodiments. Device **100A** may be included in a microprocessor, memory cell, or other integrated circuit. It will be recognized that the view of device **100A** in FIG. 1A is shown for illustration purposes and may not be drawn to scale.

(24) Device **100A** may be formed on a substrate **102** and may include fin field effect transistors (FETs) **104** and **106** as shown in FIG. 1A. Device **100A** may further include shallow trench isolation (STI) regions **108**, gate structure **110**, and spacers **112** disposed on opposite sides of gate structure **110**.

(25) In some embodiments, finFET **104** may be a multi-fin finFET having a plurality of fin structures **114** and finFET **106** may be a single-fin finFET having a fin structure **116**. Even though

FIG. 1A shows one or more multi-fin finFET **104** and one single-fin finFET **106**, device **100A** may have one or more multi-fin finFETs similar to finFET **104** and may have one or more single-fin finFETs similar to finFET **106**. In some embodiments, multi-fin finFETs such as, for example, finFET **104** may be used for high current drive devices (e.g., current sources) because of their larger effective channel width compared to single-fin finFETs such as, for example, finFET **106**. In some embodiments, single-fin finFETs such as, finFET **106** may be used for high density devices (e.g., high density memory devices) because of their smaller device area compared to multi-fin finFETs such as, for example, finFET **104**.

(26) In some embodiments, fin structures of multi-fin finFETs of device **100A** may have a smaller height compared to height of fin structures of single-fin finFETs of device **100A**. For example, each of fin structures **114** may have a height $H_{sub.1}$ shorter than height $H_{sub.2}$ of fin structure **116**, according to some embodiments. In some embodiments, height $H_{sub.1}$ may range from about 20 nm to about 40 nm and height $H_{sub.2}$ may range from about 50 nm to about 60 nm. In some embodiments, a difference between heights $H_{sub.1}$ and $H_{sub.2}$ may range from about 20 nm to about 50 nm. In some embodiments, finFET **104** may have fin-to-fin pitch $P_{sub.1}$ ranging from about 18 nm to about 24 nm.

(27) The height $H_{sub.1}$ and fin-to-fin pitch $P_{sub.1}$ of finFET **104** may be selected such that the processing steps shared to simultaneously form one or more components (e.g., STI regions **108**, polysilicon structure, gate structure **110**) of finFETs **104** and **106** is suitable for processing in high aspect ratio space between adjacent fin structures **114**. For example, in some embodiments, the height $H_{sub.1}$ and fin-to-fin pitch $P_{sub.1}$ of finFET **104** may be selected such that the shared processing steps (e.g., deposition, etching) for forming STI regions **108** and/or gate structure **110** of finFETs **104** and **106** are suitable for forming portions of STI regions **108** and/or portions of gate structure **110** in the high aspect ratio space between fin structures **114**.

(28) Substrate **102** may be a physical material on which finFETs **104** and **106** are formed. Substrate **102** may be a semiconductor material such as, but not limited to, silicon. In some embodiments, substrate **102** includes a crystalline silicon substrate (e.g., wafer). In some embodiments, substrate **102** includes (i) an elementary semiconductor, such as germanium; (ii) a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; (iii) an alloy semiconductor including silicon germanium carbide, silicon germanium, gallium arsenic phosphide, gallium indium phosphide, gallium indium arsenide, gallium indium arsenic phosphide, aluminum indium arsenide, and/or aluminum gallium arsenide; or (iv) a combination thereof. Further, substrate **102** may be doped depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, substrate **102** may be doped with p-type dopants (e.g., boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic).

(29) STI regions **108** may provide electrical isolation to finFETs **104** and **106** from each other and from neighboring active and passive elements (not illustrated herein) integrated with or deposited onto substrate **102**. STI regions **108** may be made of a dielectric material. In some embodiments, STI regions **108** may include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable insulating material. In some embodiments, STI regions **108** may include a multi-layered structure.

(30) Fin structures **114** and **116** may traverse along a Y-axis and through gate structure **110**. Portions of fin structures **114** and **116** extending above STI regions **108** may be wrapped around by gate structure **110**. In some embodiments, fin structures **114** and **116** may include material similar to substrate **102**. In some embodiments, fin structures **114** and **116** may be formed from a photolithographic patterning and an etching of substrate **102**. Fin structures **114** and **116** may have respective widths $W_{sub.1}$ and $W_{sub.2}$ in a range from about 5 nm to about 10 nm, according to some embodiments. In some embodiments, widths $W_{sub.1}$ and $W_{sub.2}$ may be equal to or different from each other. Based on the disclosure herein, it will be recognized that other widths

and materials for fin structures **114** and **116** are within the scope and spirit of this disclosure.

(31) In some embodiments, epitaxial regions **118** and **120** may be grown on portions of respective fin structures **114** and **116** that extend above STI regions **108** and are not underlying gate structure **110**, as illustrated in FIG. **1A**. Epitaxial regions **118** and **120** may include an epitaxially-grown semiconductor material. In some embodiments, the epitaxially grown semiconductor material is the same material as the material of substrate **102**. In some embodiments, the epitaxially-grown semiconductor material includes a different material from the material of substrate **102**. The epitaxially-grown semiconductor material may include: (i) a semiconductor material such as, for example, germanium or silicon; (ii) a compound semiconductor material such as, for example, gallium arsenide and/or aluminum gallium arsenide; or (iii) a semiconductor alloy such as, for example, silicon germanium and/or gallium arsenide phosphide. In some embodiments, epitaxial regions **118** and **120** may each have a thickness in a range from about 5 nm to about 15 nm around respective portions of fin structures **114** and **116** above STI regions **108**.

(32) In some embodiments, epitaxial regions **118** and **120** may be grown by (i) chemical vapor deposition (CVD) such as, for example, by low pressure CVD (LPCVD), atomic layer CVD (ALCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), or any suitable CVD; (ii) molecular beam epitaxy (MBE) processes; (iii) any suitable epitaxial process; or (iv) a combination thereof. In some embodiments, epitaxial regions **118** and **120** may be grown by an epitaxial deposition/partial etch process, which repeats the epitaxial deposition/partial etch process at least once. Such repeated deposition/partial etch process is also called a “cyclic deposition-etch (CDE) process.” In some embodiments, epitaxial regions **118** and **120** may be grown by selective epitaxial growth (SEG), where an etching gas is added to promote the selective growth of semiconductor material on the exposed surfaces of fin structures **114** and **116**, but not on insulating material (e.g., dielectric material of STI regions **108**).

(33) In some embodiments, both epitaxial regions **118** and **120** may be p-type or n-type. In some embodiments, epitaxial regions **118** and **120** may be of opposite doping type with respect to each other. In some embodiments, p-type epitaxial regions **118** and **120** may include SiGe and may be in-situ doped during an epitaxial growth process using p-type dopants such as, for example, boron, indium, or gallium. For p-type in-situ doping, p-type doping precursors such as, but not limited to, diborane (B₂H₆), boron trifluoride (BF₃), and/or other p-type doping precursors can be used.

(34) In some embodiments, each of p-type epitaxial regions **118** and **120** may have a plurality of sub-regions (not shown) that may include SiGe and may differ from each other based on, for example, doping concentration, epitaxial growth process conditions, and/or relative concentration of Ge with respect to Si. In some embodiments, each of the sub-regions may have thicknesses similar to or different from each other and thicknesses may range from about 0.5 nm to about 5 nm. In some embodiments, the atomic percent Ge in sub-regions closest to a top surface of fin structures **114** and **116** may be smaller than the atomic percent Ge in sub-regions farthest from the top surface of fin structures **114** and **116**. In some embodiments, the sub-regions closest to the top surface of fin structures **114** and **116** may include Ge in a range from about 15 atomic percent to about 35 atomic percent, while the sub-regions farthest from the top surface of fin structures **114** and **116** may include Ge in a range from about 25 atomic percent to about 50 atomic percent with any remaining atomic percent being Si in the sub-regions.

(35) The plurality of sub-regions of p-type epitaxial regions **118** and **120** may be epitaxially grown under a pressure of about 10 Torr to about 300 Torr and at a temperature of about 500° C. to about 700° C. using reaction gases such as HCl as an etching agent, GeH₄ as Ge precursor, dichlorosilane (DCS) and/or SiH₄ as Si precursor, B₂H₆ as B dopant precursor, H₂, and/or N₂. To achieve different concentration of Ge in the plurality of sub-regions, the ratio of a flow rate of Ge to Si precursors may be varied during their respective growth process, according to some embodiments. For example, a Ge to Si precursor flow rate ratio in a range from about 9 to about 25 may be used

during the epitaxial growth of the sub-regions closest to the top surface of fin structures **114** and **116**, while a Ge to Si precursor flow rate ratio less than about 6 may be used during the epitaxial growth of the sub-regions farthest from the top surface of fin structures **114** and **116**.

(36) The plurality of sub-regions of p-type epitaxial regions **118** and **120** may have varying p-type dopant concentration with respect to each other, according to some embodiments. For example, the sub-regions closest to the top surface of fin structures **114** and **116** may be undoped or may have a dopant concentration lower (e.g., dopant concentration less than about 8×10^{20} atoms/cm³) than the dopant concentrations (e.g., dopant concentration in a range from about 1×10^{20} to about 3×10^{22} atoms/cm³) of the sub-regions farthest from the top surface of fin structures **114** and **116**.

(37) In some embodiments, n-type epitaxial regions **118** and **120** may include Si and may be in-situ doped during an epitaxial growth process using n-type dopants such as, for example, phosphorus or arsenic. For n-type in-situ doping, n-type doping precursors such as, but not limited to, phosphine (PH₃), arsine (AsH₃), and/or other n-type doping precursor can be used. In some embodiments, each of n-type epitaxial regions **118** and **120** may have a plurality of n-type sub-regions. Except for the type of dopants, the plurality of n-type sub-regions may be similar to the plurality of p-type sub-regions, in thickness, relative Ge concentration with respect to Si, dopant concentration, and/or epitaxial growth process conditions.

(38) Based on the disclosure herein, it will be recognized that other materials, thicknesses, Ge concentrations, and dopant concentrations for the plurality of n-type and/or p-type sub-regions are within the scope and spirit of this disclosure.

(39) Fin structures **114** and **116** are current-carrying structures for respective finFETs **104** and **106**. Epitaxial regions **118** and **120** along with the portions of fin structures **114** and **116** covered by respective epitaxial regions **114** and **116** are configured to function as source/drain (S/D) regions of respective finFETs **104** and **106**. Channel regions (not shown) of finFETs **104** and **106** may be formed in portions of their respective fin structures **114** and **116** underlying gate structure **110**.

(40) Gate structure **110** may include a dielectric layer **122** and a gate electrode **124**. Additionally, in some embodiments, gate structure **110** may include another dielectric layer **125**. Gate structure **110** may have a horizontal dimension (e.g., gate length) L_g that ranges from about 5 nm to about 30 nm, according to some embodiments. Gate structure **110** may be formed by a gate replacement process.

(41) In some embodiments, dielectric layer **122** is adjacent to and in contact with gate electrode **124**. Dielectric layer **122** may have a thickness $122t$ in a range of about 1 nm to about 5 nm. Dielectric layer **122** may include silicon oxide and may be formed by CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), e-beam evaporation, or other suitable process. In some embodiments, dielectric layer **122** may include (i) a layer of silicon oxide, silicon nitride, and/or silicon oxynitride, (ii) a high-k dielectric material such as, for example, hafnium oxide (HfO₂), titanium oxide (TiO₂), hafnium zirconium oxide (HfZrO), tantalum oxide (Ta₂O₃), hafnium silicate (HfSiO₄), zirconium oxide (ZrO₂), zirconium silicate (ZrSiO₂), (iii) a high-k dielectric material having oxides of lithium (Li), beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), scandium (Sc), yttrium (Y), zirconium (Zr), aluminum (Al), lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), or lutetium (Lu), or (iv) a combination thereof. High-k dielectric layers may be formed by ALD and/or other suitable methods. In some embodiments, dielectric layer **122** may include a single layer or a stack of insulating material layers. Based on the disclosure herein, it will be recognized that other materials and formation methods for dielectric layer **122** are within the scope and spirit of this disclosure.

(42) In some embodiments, dielectric layer **125** may be formed as an interlayer between STI regions **108** and spacers **112** and between STI regions **108** and gate structure **110**. Dielectric layer

125 may have a composition similar to dielectric layer **122**. In some embodiments, dielectric layers **122** and **125** may function as gate dielectric layers of gate structure **110**. In some embodiments, dielectric layer **125** may have a thickness smaller than thickness **122t** of dielectric layer **122**.

(43) Gate electrode **124** may include a gate work function metal layer (not shown) and a gate metal fill layer (not shown). In some embodiments, gate work function metal layer is disposed on dielectric layer **122**. The gate work function metal layer may include a single metal layer or a stack of metal layers. The stack of metal layers may include metals having work functions similar to or different from each other. In some embodiments, the gate work function metal layer may include, for example, aluminum (Al), copper (Cu), tungsten (W), titanium (Ti), tantalum (Ta), titanium nitride (TiN), tantalum nitride (TaN), nickel silicide (NiSi), cobalt silicide (CoSi), silver (Ag), tantalum carbide (TaC), tantalum silicon nitride (TaSiN), tantalum carbon nitride (TaCN), titanium aluminum (TiAl), titanium aluminum nitride (TiAlN), tungsten nitride (WN), metal alloys, and/or combinations thereof. The gate work function metal layer may be formed using a suitable process such as ALD, CVD, PVD, plating, or combinations thereof. In some embodiments, the gate work function metal layer has a thickness in a range from about 2 nm to about 15 nm. Based on the disclosure herein, it will be recognized that other materials, formation methods, and thicknesses for the gate work function metal layer are within the scope and spirit of this disclosure.

(44) The gate metal fill layer may include a single metal layer or a stack of metal layers. The stack of metal layers may include metals different from each other. In some embodiments, the gate metal fill layer may include a suitable conductive material such as, for example, Ti, silver (Ag), Al, titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbo-nitride (TaCN), tantalum silicon nitride (TaSiN), manganese (Mn), Zr, titanium nitride (TiN), tantalum nitride (TaN), ruthenium (Ru), molybdenum (Mo), tungsten nitride (WN), copper (Cu), tungsten (W), cobalt (Co), nickel (Ni), titanium carbide (TiC), titanium aluminum carbide (TiAlC), tantalum aluminum carbide (TaAlC), metal alloys, and/or combinations thereof. The gate metal fill layer may be formed by ALD, PVD, CVD, or other suitable deposition process. Based on the disclosure herein, it will be recognized that other materials and formation methods for the gate metal fill layer are within the scope and spirit of this disclosure.

(45) Spacers **112** may form sidewalls of gate structure **110** and are in contact with dielectric layer **122**. Spacers **112** may include insulating material such as, for example, silicon oxide, silicon nitride, a low-k material, or a combination thereof. Spacers **112** may have a low-k material with a dielectric constant less than 3.9 (e.g., less than 3.5, 3, or 2.8). In some embodiments, each of spacers **112** may have a thickness **112t** in a range from about 7 nm to about 10 nm. Based on the disclosure herein, it will be recognized that other materials and thicknesses for spacers **112** are within the scope and spirit of this disclosure.

(46) Referring back to FIG. **1A**, device **100A** may further include etch stop layer (ESL) **126**, interlayer dielectric (ILD) **128**, and source/drain (S/D) contact structures **130** and **132** of respective finFETs **104** and **106**, according to some embodiments.

(47) ESL **126** may be configured to protect gate structure **110** and/or portions of epitaxial regions **118** and **120** that are not in contact with source/drain (S/D) contact structures **130** and **132**. This protection may be provided, for example, during formation of ILD layer **128** and/or S/D contact structures **130** and **132**. ESL **126** may be disposed on sides of spacers **112**. In some embodiments, ESL **126** may include, for example, silicon nitride (SiN.sub.x), silicon oxide (SiO.sub.x), silicon oxynitride (SiON), silicon carbide (SiC), silicon carbo-nitride (SiCN), boron nitride (BN), silicon boron nitride (SiBN), silicon carbon boron nitride (SiCBN), or a combination thereof. In some embodiments, ESL **126** may include silicon nitride or silicon oxide formed by low pressure chemical vapor deposition (LPCVD), plasma enhanced chemical vapor deposition (PECVD), chemical vapor deposition (CVD), or silicon oxide formed by a high-aspect-ratio process (HARP). In some embodiments, ESL **126** has a thickness **126t** in a range from about 3 nm to 10 nm or from about 10 nm to about 30 nm. Based on the disclosure herein, it will be recognized that other

materials, formation methods, and thicknesses for ESL **126** are within the scope and spirit of this disclosure.

(48) ILD layer **128** may be disposed on ESL **126** and may include a dielectric material deposited using a deposition method suitable for flowable dielectric materials (e.g., flowable silicon oxide, flowable silicon nitride, flowable silicon oxynitride, flowable silicon carbide, or flowable silicon oxycarbide). For example, flowable silicon oxide may be deposited using flowable CVD (FCVD). In some embodiments, the dielectric material is silicon oxide. In some embodiments, ILD layer **128** may have a thickness **128t** in a range from about 50 nm to about 200 nm. Based on the disclosure herein, it will be recognized that other materials, thicknesses, and formation methods for ILD layer **128** are within the scope and spirit of this disclosure.

(49) S/D contact structures **130** and **132** may be configured to electrically connect respective S/D regions of finFETs **104** and **106** to other elements of device **100A** and/or of the integrated circuit. S/D contact structures **130** and **132** may be formed within ILD layer **128**. S/D contact structure **130** may include a metal silicide layer **134** and a conductive region **136** over metal silicide layer **134**, and S/D contact structure **132** may include a metal silicide layer **138** and a conductive region **140** over metal silicide layer **138**. In some embodiments, there may be conductive liners (not shown) between metal silicide layer **134** and conductive region **136** and between metal silicide layer **138** and conductive region **140**. The conductive liners may be configured as diffusion barriers to prevent diffusion of unwanted atoms and/or ions into S/D regions of finFETs **104** and **106** during formation of conductive regions **136** and **140**. In some embodiments, the conductive liners may include a single layer or a stack of conductive materials such as, for example, TiN, Ti, Ni, TaN, Ta, or a combination thereof. In some embodiments, the conductive liners may act as an adhesion-promoting-layer, a glue-layer, a primer-layer, a protective-layer, and/or a nucleation-layer. The conductive liners may have a thickness in a range from about 1 nm to about 2 nm, according to some embodiments.

(50) In some embodiments, silicide layers **134** and **138** may include metal silicides and may provide a low resistance interface between respective conductive regions **136** and **140** and corresponding S/D regions of finFETs **104** and **106**. Examples of metal used for forming the metal silicides are Co, Ti, or Ni.

(51) In some embodiments, conductive regions **136** and **140** may include conductive materials such as, for example, W, Al, or Co. In some embodiments, conductive regions **136** and **140** may each have an average horizontal dimension (e.g., width) in a range from about 15 nm to about 25 nm and may each have an average vertical dimension (e.g., height) in a range from about 400 nm to about 600 nm. Based on the disclosure herein, it will be recognized that other materials and dimensions for conductive liners, silicide layers **134** and **138**, and conductive regions **136** and **140** are within the scope and spirit of this disclosure.

(52) FIG. **1B** is an isometric view of a device **100B**, according to some embodiments. Elements in FIG. **1B** with the same annotations as elements in FIG. **1A** are described above. Device **100B** may be included in a microprocessor, memory cell, or other integrated circuit. It will be recognized that the view of device **100B** in FIG. **1B** is shown for illustration purposes and may not be drawn to scale.

(53) Device **100B** may be formed on a substrate **102** and may include finFETs **104** and **106*** as shown in FIG. **1B**. Device **100A** may further include shallow trench isolation (STI) regions **108**, gate structure **110**, spacers **112** disposed on opposite sides of gate structure **110**, ESL **126**, ILD layer **128**, and contact structures **130** and **132***. The above discussion of finFET **106** and contact structure **132*** applies to respective finFET **106*** and contact structure **132*** unless mentioned otherwise.

(54) In some embodiments, finFET **104** may be a multi-fin finFET having a plurality of fin structures **114** and finFET **106*** may be a multi-fin finFET having fin structures **116**. Even though FIG. **1B** shows one multi-fin finFET **104** and one multi-fin finFET **106***, device **100B** may have

one or more multi-fin finFETs similar to finFETs **104** and **106***. In some embodiments, each of fin structures **114** of finFET **104** may have a smaller height H.sub.1 compared to a height H.sub.2 of each of fin structures **116** of finFET **106*** of device **100B**. In some embodiments, height H.sub.1 may range from about 20 nm to about 40 nm and height H.sub.2 may range from about 50 nm to about 60 nm. In some embodiments, a difference between heights H.sub.1 and H.sub.2 may range from about 20 nm to about 50 nm. In some embodiments, a fin-to-fin pitch P.sub.1 of finFET **104** may be smaller compared to a fin-to-fin pitch P.sub.2 of finFET **106***. In some embodiments, fin-to-fin pitch P.sub.1 may range from about 18 nm to about 24 nm and fin-to-fin pitch P.sub.2 may range from about 24 nm to about 34 nm.

(55) The heights and fin-to-fin pitches of finFETs **104** and **106*** may be selected such that the processing steps shared to simultaneously form one or more components (e.g., STI regions **108**, polysilicon structure, gate structure **110**) of finFETs **104** and **106*** is suitable for processing in high aspect ratio space between adjacent fin structures **114** of finFET **104** and between adjacent fin structures **116** of finFET **106***.

(56) FIG. 2A is an isometric view of a device **200A**, according to some embodiments. Elements in FIG. 2A with the same annotations as elements in FIG. 1A are described above. Device **200A** may be included in a microprocessor, memory cell, or other integrated circuit. It will be recognized that the view of device **200A** in FIG. 2A is shown for illustration purposes and may not be drawn to scale.

(57) Device **200A** may be formed on a substrate **102** and may include finFETs **204** and **206** as shown in FIG. 2A. Device **200A** may further include shallow trench isolation (STI) regions **108**, gate structure **110**, spacers **112** disposed on opposite sides of gate structure **110**, ESL **126**, ILD layer **128**, and contact structures **130** and **13**. The above discussion of finFETs **104** and **106** applies to respective finFETs **204** and **206** unless mentioned otherwise.

(58) FinFET **204** may include fin structures **214** and epitaxial source/drain (S/D) regions **218** and finFET **206** may include fin structure **216** and epitaxial S/D region **220**. The above discussion of fin structures **114** and **116** applies to fin structures **214** and **216** and the discussion of epitaxial regions **118** and **120** applies to epitaxial S/D regions **218** and **220** unless mentioned otherwise. Fin structures **214** and **216** may traverse along a Y-axis and through gate structure **110**.

(59) In some embodiments, S/D regions **218** and **220** may be epitaxially formed from top surface of fin structures **214** and **216** after an etch back process performed on portions of fin structures **214** and **216** that are not underlying gate structure **110**. S/D regions **218** and **220** may form respective interfaces **215** and **217** with fin structures **214** and **216**. In some embodiments, interfaces **215** and **217** are on the same plane as top surface of STI regions **108**. In some embodiments, interfaces **215** and **217** are either above or below the level of interface **109** formed between STI regions **108** and substrate **102**.

(60) In some embodiments, epitaxial S/D regions **218** of finFET **204** may be unmerged as shown in FIG. 2A. Additionally or alternatively to fin structures **218**, finFET **204** may have merged epitaxial S/D region **218*** as shown in FIG. 2B. FIG. 2B is an isometric view of a device **200B**, according to some embodiments. Elements in FIG. 2B with the same annotations as elements in FIGS. 1A and 2A are described above. The above discussion of epitaxial S/D regions **218** applies to epitaxial S/D region **218*** unless mentioned otherwise.

(61) FIGS. 1A-1B and 2A-2B show one gate structure **110**. However, based on the disclosure herein, it will be recognized that devices **100A**, **100B**, **200A**, and/or **200B** may have additional gate structures similar and parallel to gate structure **110**. In addition, device **100A**, **100B**, **200A**, and/or **200B** may be incorporated into an integrated circuit through the use of other structural components such as gate contact structures, conductive vias, conductive lines, dielectric layers, passivation layers, etc., that are omitted for the sake of clarity. Based on the disclosure herein, it will be recognized that cross-sectional shapes of STI regions **108**, spacers **112**, fin structures **114**, **116**, **214**, and **216**, and epitaxial regions **118**, **120**, **218**, **220**, and **218*** are illustrative and are not intended to

be limiting.

(62) FIG. 3 is a flow diagram of an example method **300** for fabricating devices **100A** and/or **100B**, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. 3 will be described with reference to the example fabrication process for fabricating devices **100A** and **100B** as illustrated in FIGS. 4A-13A and FIGS. 4B-13B, respectively. FIGS. 4A-13A and 4A-13B are isometric views of respective devices **100A** and **100B** at various stages of their fabrication, according to some embodiments. Operations can be performed in a different order or not performed depending on specific applications. It should be noted that method **300** does not produce complete devices **100A** and **100B**. Accordingly, it is understood that additional processes may be provided before, during, and after method **300**, and that some other processes may only be briefly described herein. Elements in FIGS. 4A-13A and 4B-13B with the same annotations as elements in FIGS. 1A-1B and 2A-2B are described above.

(63) In operation **305**, fin structures of first and second finFETs are formed on a substrate. For example, as shown in FIGS. 4A and 4B, fin structures **114** of finFET **104** and fin structures **116** of finFETs **106** and **106*** are formed on substrate **102**. Fin structures **114** and **116** may be formed by etching substrate **102** through patterned hard mask layers **442** and **444** formed on unetched substrate **102**. In some embodiments, hard mask layer **442** may be a thin film including silicon oxide formed, for example, using a thermal oxidation process. In some embodiments, hard mask layer **444** may be formed of silicon nitride using, for example, low pressure chemical vapor deposition (LPCVD) or plasma enhanced CVD (PECVD). In some embodiments, fin structures **114** and **116** each may have fin widths W less than about 30 nm.

(64) In referring to FIG. 3, in operation **310**, a layer of insulating material for STI regions is deposited and the patterned hard mask layers are removed. For example, a layer of insulating material **108*** may be blanket deposited on the structures of FIGS. 4A and 4B followed by a chemical mechanical polishing (CMP) process to form the structures of FIGS. 5A and 5B. The CMP process may remove the patterned hard mask layers **442** and **444** and portions of layer of insulating material **108*** to substantially coplanarize a top surface of layer of insulating material **108*** with top surfaces of fin structures **114** and **116** as shown in FIGS. 5A and 5B.

(65) In some embodiments, layer of insulating material **108*** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), or a low-k dielectric material. In some embodiments, layer of insulating material **108*** may be deposited using a flowable chemical vapor deposition (FCVD) process, a high-density-plasma (HDP) CVD process, using silane (SiH_4) and oxygen (O_2) as reacting precursors. In some embodiments, layer of insulating material **108*** may be formed using a sub-atmospheric CVD (SACVD) process or high aspect-ratio process (HARP), where process gases may include tetraethoxysilane (TEOS) and/or ozone (O_3). In some embodiments, layer of insulating material **108*** may be formed using a spin-on-dielectric (SOD) such as, for example, hydrogen silsesquioxane (HSQ) or methyl silsesquioxane (MSQ).

(66) In referring to FIG. 3, in operation **315**, a hard mask layer is formed on the fin structure of the second finFET and a portion of the fin structure of the first finFET is etched back. For example, a layer of insulating material may be blanket deposited on the structures of FIGS. 5A and 5B and then patterned using photolithography and a dry etching process (e.g., reaction ion etching process) to form thin hard mask layer **646** on finFETs **106** and **106*** as shown in respective FIGS. 6A and 6B. Thin hard mask layer **646** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, or a low-k dielectric material. In some embodiments, thin hard mask layer **646** may be deposited using CVD, ALD, HDP CVD process, or a suitable process for depositing a thin layer of insulating material. In some embodiments, thin hard mask layer **646** may have a thickness ranging from about 2 nm to about 8 nm (e.g., about 3 nm, about 5 nm, or about 7 nm). Based on the disclosure herein, it will be recognized that other thicknesses and materials for thin hard mask layer **646** are within the scope and spirit of this disclosure.

(67) The formation of thin hard mask layer **646** may be followed by an etch back process of fin structures **114** of finFET **104** to form recessed regions **650** within layer of insulating material **108***. The fin structures **114** may be etched back by a vertical dimension **648** from a top surface of layer of insulating material **108***. In some embodiments, vertical dimension **648** may range from about 20 nm to about 50 nm. The etch back process may include a dry etching process (e.g., reaction ion etching process using a chlorine based etchant). Thin hard mask layer **646** may prevent fin structures **116** from being etched back during the etch back process of fin structures **114**, and consequently, facilitate the formation of fin structures of different heights, such as, for example, height H.sub.1 of fin structures **114** ranging from about 20 nm to about 40 nm and height H.sub.2 of fin structures **116** ranging from about 50 nm to about 60 nm. In some embodiments, during the etch back process of fin structures **114**, the thickness of thin hard mask layer **646** may be reduced to a thickness **646t** ranging from about 1 nm to about 3 nm.

(68) In referring to FIG. 3, in operation **320**, a masking region is formed on the etched back fin structure of the first finFET. For example, a layer of insulating material may be blanket deposited on the structures of FIGS. 6A and 6B followed by a CMP process to form masking regions **752** as shown in respective FIGS. 7A and 7B. The CMP process may be performed until top surfaces of masking regions **752**, layer of insulating material **108***, and fin structures **116** are substantially coplanar. In some embodiments, top surfaces of layer of insulating material **108*** and fin structures **116** may act as CMP stop layer. In some embodiments, dry and/or wet etch processes may be used instead of or in combination with the CMP process to form masking regions **752**. Masking regions **752** may include, for example, silicon oxide, silicon nitride, silicon oxynitride, a low-k dielectric material, or a suitable insulating material. In some embodiments, the layer of insulating material for forming masking regions **752** may be deposited using CVD, ALD, HDP CVD process, or a suitable process for depositing a layer of insulating material. Based on the disclosure herein, it will be recognized that other materials for masking regions **752** are within the scope and spirit of this disclosure.

(69) In referring to FIG. 3, in operation **325**, STI regions are formed. For example, as shown in FIGS. 8A and 8B, STI regions **108** are formed. STI regions **108** may be formed by etching back layer of insulating material **108*** of the structures shown in FIGS. 7A and 7B. In some embodiments, masking regions **752** may be etched during the etch back of layer of insulating material **108***. The etchants used to etch back layer of insulating material **108*** may have similar etch selectivity to masking regions **752** and layer of insulating material **108***.

(70) The etch back of layer of insulating material **108*** may be performed, for example, by a dry etch process, a wet etch process, or a combination thereof. In some embodiments, the dry etch process may include using a plasma dry etch with a gas mixture having octafluorocyclobutane (C.sub.4F.sub.8), argon (Ar), oxygen (O.sub.2), and helium (He), fluoroform (CHF.sub.3) and He, carbon tetrafluoride (CF.sub.4), difluoromethane (CH.sub.2F.sub.2), chlorine (Cl.sub.2), and O.sub.2, hydrogen bromide (HBr), O.sub.2, and He, or a combination thereof with a pressure ranging from about 1 mTorr to about 5 mTorr. In some embodiments, the wet etch process may include using a diluted hydrofluoric acid (DHF) treatment, an ammonium peroxide mixture (APM), a sulfuric peroxide mixture (SPM), hot deionized water (DI water), or a combination thereof. In some embodiments, the wet etch process may include using an etch process that may use ammonia (NH.sub.3) and hydrofluoric acid (HF) as etchants and inert gases such as, for example, Ar, xenon (Xe), He, or a combination thereof. In some embodiments, the flow rate of HF and NH.sub.3 used in the etch process may each range from about 10 sccm to about 100 sccm (e.g., about 20 sccm, 30 sccm, or 40 sccm). In some embodiments, the etch process may be performed at a pressure ranging from about 5 mTorr to about 100 mTorr (e.g., about 20 mTorr, about 30 mTorr, or about 40 mTorr) and a high temperature ranging from about 50° C. to about 120° C.

(71) In referring to FIG. 3, in operation **330**, a dielectric layer is deposited. For example, as shown in FIGS. 9A and 9B, a dielectric layer **125*** may be blanket deposited on the structures of FIGS. 8A

and **8B**. Dielectric layer **125*** may form dielectric layer **125** (shown in FIGS. **1A-1B** and **2A-2B**) in subsequent processing. Dielectric layer **125*** may include a suitable dielectric material, such as, for example, silicon oxide and may be deposited using a suitable dielectric material deposition process, such as, for example, CVD or ALD.

(72) In referring to FIG. **3**, in operation **335**, a polysilicon structure and epitaxial regions are formed on the fin structures of the first and second finFET. For example, polysilicon structure **1056** and epitaxial regions **118** and **120** may be formed as shown in FIGS. **10A** and **10B**. Polysilicon structure **1056** may be formed on the structures of FIGS. **9A** and **9B**. In some embodiments, a vertical dimension **1056t** of polysilicon structure **1056** may be in a range from about 90 nm to about 200 nm. In some embodiments, polysilicon structure **760** and hard mask layers **1058** and **1060** may be replaced in a gate replacement process during subsequent processing to form gate structure **110** discussed above.

(73) In some embodiments, polysilicon structure **1056** may be formed by blanket deposition of polysilicon, followed by photolithography and etching of the deposited polysilicon. The deposition process may include chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), other suitable deposition methods, or a combination thereof.

Photolithography may include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), other suitable processes, or a combination thereof. Etching processes may include dry etching, wet etching, and/or other etching methods (e.g., reactive ion etching).

(74) In some embodiments, hard mask layers **1058** and **1060** may be patterned on polysilicon structure **1056** to protect polysilicon structure **1056** from subsequent processing steps. Hard mask layers **1058** and **1060** may include insulating material such as, for example, silicon nitride.

(75) The formation of hard mask layers **1058** and **1060** may be followed by formation of spacers **112** on sidewalls of polysilicon structure **1056**. Spacers **112** may be selectively formed on sidewalls of polysilicon structure **1056** and may not be formed on dielectric layer **125*** of FIGS. **9A** and **9B**. The selective formation of spacers **112** may include a surface treatment and a deposition process. The surface treatment may include exposing dielectric layer **125*** and polysilicon structure **1056** to an inhibitor to form an inhibiting layer (not shown) on top surface of dielectric layer **125*** and to form a H- or F-terminated surfaces on the sidewalls of polysilicon structure **1056**. The inhibiting layer may have a hydroxyl-terminated surface. The H- or F-terminated surfaces may facilitate the deposition of the material of spacers **112**. The surface treatment may further include selectively converting the hydroxyl-terminated surface to a hydrophobic surface by including a hydrophobic component (e.g., a component having carbon) to the hydroxyl-terminated surface. In some embodiments, an etching process performed at about 45° C. can be used to remove native oxide from the hydroxyl-terminated surface to convert the hydroxyl-terminated surface to a hydrophobic surface. In some embodiments, the etching process is performed using process gases such as, for example, nitrogen trifluoride, ammonia, hydrogen fluoride, other suitable gas, and/or combinations thereof. In some embodiments, the etching process is performed using a combined gas of nitrogen trifluoride and hydrogen. In some embodiments, the etching process is performed using a combined gas of hydrogen fluoride and ammonia. The hydrophobic surface may prevent deposition of the material of spacers **112** on dielectric layer **125***. The surface treatment may be followed by the deposition of the material of spacer **112**.

(76) In some embodiments, the material of spacers **112** may be deposited using, for example, CVD or ALD. The surface treatment may be performed before or during the deposition process. The deposition process may be followed by, for example, an oxygen plasma treatment to remove the hydrophobic component and the inhibitor layer on the top surface of dielectric layer **125***. In some embodiments, spacer **112** may include (i) a dielectric material such as, for example, silicon oxide, silicon carbide, silicon nitride, silicon oxy-nitride, (ii) an oxide material, (iii) a nitride material, (iv) a low-k material, or (v) a combination thereof. In some embodiments, dielectric layer **125***

may include silicon oxide and spacers **112** may include silicon nitride.

(77) The selective formation of spacers **112** may followed by formation of dielectric layer **125** (shown in FIGS. **10A** and **10B**) by etching of dielectric layer **125*** from regions that are not covered by polysilicon structure **1056** and spacers **112**. The etch process may include a wet etch process using, for example, diluted HF. This etch process may etch native oxide from top surfaces of fin structures **114** and **116**. In some embodiments, this etch process may etch some portions of STI regions **108** and consequently, form curved top surfaces **108s** of STI regions **108**.

(78) The etching of dielectric layer **125*** may be followed by the growth of epitaxial regions **118** and **120** on respective fin structures **114** and **116**. In some embodiments, epitaxial regions **118** and **120** may be grown by (i) chemical vapor deposition (CVD) such as, for example, by low pressure CVD (LPCVD), atomic layer CVD (ALCVD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), or any suitable CVD; (ii) molecular beam epitaxy (MBE) processes; (iii) any suitable epitaxial process; or (iv) a combination thereof. In some embodiments, epitaxial regions **118** and **120** may be grown by an epitaxial deposition/partial etch process, which repeats the epitaxial deposition/partial etch process at least once. Such repeated deposition/partial etch process is also called a “cyclic deposition-etch (CDE) process.” In some embodiments, epitaxial regions **118** and **120** may be grown by selective epitaxial growth (SEG), where an etching gas is added to promote the selective growth of semiconductor material on the exposed surfaces of fin structures **114** and **116**, but not on insulating material (e.g., dielectric material of STI regions **108**).

(79) In some embodiments, both epitaxial regions **118** and **120** may be p-type or n-type. In some embodiments, epitaxial regions **118** and **120** may be of opposite doping type with respect to each other. In some embodiments, p-type epitaxial regions **118** and **120** may include SiGe and may be in-situ doped during an epitaxial growth process using p-type dopants such as, for example, boron, indium, or gallium. For p-type in-situ doping, p-type doping precursors such as, but not limited to, diborane (B₂H₆), boron trifluoride (BF₃), and/or other p-type doping precursors can be used. In some embodiments, n-type epitaxial regions **118** and **120** may include Si and may be in-situ doped during an epitaxial growth process using n-type dopants such as, for example, phosphorus or arsenic. For n-type in-situ doping, n-type doping precursors such as, but not limited to, phosphine (PH₃), arsine (AsH₃), and/or other n-type doping precursor can be used.

(80) In some embodiments, instead of the growth of epitaxial regions **118** and **120**, the etching of dielectric layer **125*** may be followed by etch back of fin structures **114** and **116** to form fin structures **214** and **216** as discussed above with reference to FIGS. **2A** and **2B**. The formation of fin structures **214** and **216** may be followed by the epitaxial growth of S/D regions **218**, **220**, and **218*** as discussed above.

(81) In referring to FIG. **3**, in operation **340**, the polysilicon structure is replaced with a gate structure. For example, as shown in FIGS. **11A** and **11B** gate structure **110** may be formed after removing polysilicon structure **1056**. In some embodiments, prior to the removal of polysilicon structure **1056**, ESL **126** and ILD layer **128** may be formed as shown in FIGS. **11A** and **11B**. In some embodiments, ESL **126** may include, for example, SiN_x, SiON, SiC, SiCN, BN, SiBN, SiCBN, or a combination thereof. In some embodiments, ESL **126** may include silicon nitride formed by low pressure chemical vapor deposition (LPCVD), plasma enhanced chemical vapor deposition (PECVD), chemical vapor deposition (CVD), or atomic layer deposition (ALD). In some embodiments, ILD layer **128** may include a dielectric material. The dielectric material of ILD layer **128** may be deposited using a deposition method suitable for flowable dielectric materials (e.g., flowable silicon oxide). For example, flowable silicon oxide may be deposited for ILD layer **128** using flowable CVD (FCVD).

(82) The removal of polysilicon structure **1056** and hard mask layers **1058** and **1060** may be performed using a dry etching process (e.g., reaction ion etching) or a wet etching process. In some embodiments, the gas etchants used in etching of polysilicon structure **1056** and hard mask layers **1058** and **1060** may include chlorine, fluorine, or bromine. In some embodiments, an NH₄OH

wet etch may be used to remove polysilicon structure **1056**, or a dry etch followed by a wet etch process may be used to remove polysilicon structure **1056**.

(83) The formation of gate structure **110** may include deposition of dielectric layer **122**. Dielectric layer **122** may include silicon oxide and may be formed by CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), e-beam evaporation, or other suitable process. In some embodiments, dielectric layer **122** may include (i) a layer of silicon oxide, silicon nitride, and/or silicon oxynitride, (ii) a high-k dielectric material such as, for example, hafnium oxide (HfO₂), TiO₂, HfZrO, Ta₂O₃, HfSiO₄, ZrO₂, ZrSiO₂, (iii) a high-k dielectric material having oxides of Li, Be, Mg, Ca, Sr, Sc, Y, Zr, Al, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, or Lu, or (iv) a combination thereof. High-k dielectric layers may be formed by ALD and/or other suitable methods. In some embodiments, dielectric layer **122** may include a single layer or a stack of insulating material layers.

(84) The deposition of dielectric layer **122** may be followed by deposition of gate electrode **124**. Gate electrode **124** may include a single metal layer or a stack of metal layers. The stack of metal layers may include metals different from each other. In some embodiments, gate electrode **124** may include a suitable conductive material such as, for example, Ti, Ag, Al, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, WN, Cu, W, Co, Ni, TiC, TiAlC, TaAlC, metal alloys, and/or combinations thereof. Gate electrode **124** may be formed by ALD, PVD, CVD, or other suitable deposition process.

(85) The deposited dielectric layer **122** and gate electrode **124** may be planarized by a CMP process. The CMP process may coplanarize top surfaces of dielectric layer **122** and gate electrode **124** with top surface ILD layer **128** as shown in FIGS. **11A** and **11B**.

(86) In referring to FIG. **3**, in operation **345**, S/D contact openings are formed on the epitaxial regions. For example, as shown in FIGS. **12A** and **12B**, S/D contact openings **1262** and **1264** may be formed on respective epitaxial regions **114** and **116**. The formation of S/D contact openings **1262** and **1264** may include (i) removing portions of ILD layer **128** overlying epitaxial regions **118** and **120** and (ii) removing portions of ESL **126** underlying the etched portions of ILD layer **128**. The removal of the portions of ILD layer **128** may include patterning using photolithography to expose areas on top surface of ILD layer **128** corresponding to the portions of ILD layer **128** that are to be removed. The portions of ILD layer **128** may be removed by a dry etching process. In some embodiments, the dry etching process may be a fluorine-based process.

(87) The ILD etch process may include two steps. In the first etch step, etching may be performed using CF₄ gas at a flow rate ranging from about 50 sccm to about 500 sccm. In the second etch step, etching may be performed using a gas mixture including C₄F₆ gas at a flow rate ranging from about 5 sccm to about 50 sccm, Ar gas at a flow rate ranging from about 100 sccm to about 500 sccm, and O₂ gas at a flow rate ranging from about 5 sccm to about 50 sccm. In some embodiments, each of the first and second etch steps may be carried out for a time period ranging from about 1 sec to about 60 sec. In some embodiments, each of the first and second etch steps may be performed at a temperature ranging from about 10° C. to about 100° C., under a pressure ranging from about 3 mTorr to about 500 mTorr, and at an RF power ranging from about 300 W to about 800 W. In some embodiments, the first etch step has a higher etch rate than the second etch step.

(88) The etching of the portions of ILD layer **128** may be followed by a dry etching of portions of ESL **126** underlying the etched portions of ILD layer **128**. In some embodiments, these portions of ESL **126** may be etched in two steps. In the first etch step, etching may be performed using a gas mixture including difluoromethane (CH₂F₂) gas at a flow rate ranging from about 5 sccm to about 50 sccm and carbon tetrafluoride (CF₄) gas at a flow rate ranging from about 10 sccm to about 100 sccm. In the second etch step, etching may be performed using a gas mixture including fluoromethane (CH₃F) gas at a flow rate ranging from about 5 sccm to about 50 sccm, Ar gas at a flow rate ranging from about 100 sccm to about 500 sccm, and H₂ gas at a

flow rate ranging from about 100 sccm to about 500 sccm. In some embodiments, each of the first and second etch steps may be carried out for a time period ranging from about 1 sec to about 60 sec. In some embodiments, each of the first and second etch steps may be performed at a temperature ranging from about 10° C. to about 100° C., under a pressure ranging from about 10 mTorr to about 100 mTorr, and at an RF power ranging from about 500 W to about 800 W. In some embodiments, the first etch step has a higher etch rate than the second etch step.

(89) In some embodiments, the formation of S/D contact openings **1262** and **1264** may be followed by formation of metal silicide layers **134** and **138** as shown in FIGS. **12A** and **12B**. In some embodiments, the metal used for forming metal silicides may include Co, Ti, or Ni. In some embodiments, TiN, Ti, Ni, Co, or a combination thereof is deposited by ALD or CVD to form diffusion barrier layers (not shown) along surfaces of S/D contact openings **1262** and **1264**. This deposition of diffusion barrier layers is followed by a rapid thermal annealing process at a temperature in a range from about 700° C. to about 900° C. to form metal silicide layers **134** and **138**.

(90) In referring to FIG. **3**, in operation **350**, S/D contact structures are formed in the S/D contact openings. For example, as shown in FIGS. **13A** and **13B**, S/D contact structures **130**, **132**, and **132*** may be formed in contact openings **1262** and **1264**. The formation of conductive regions **136** and **140** of respective contact structures **130** and **132** may include deposition of materials of conductive regions **136** and **140**. Blanket deposition of the materials of conductive **136** and **140** may be performed using, for example, PVD, CVD, or ALD, on the structures of FIGS. **12A** and **12B**. In some embodiments, conductive regions **136** and **140** may include a conductive material such as, for example, W, Al, Co, Cu, or a suitable conductive material.

(91) The deposition of the materials of conductive regions **136** and **140** may be followed by a CMP process to coplanarize top surfaces of conductive regions **136** and **140** with top surface of ILD layer **128**. In some embodiments, the CMP process, may use a silicon or an aluminum abrasive with abrasive concentrations ranging from about 0.1% to about 3%. In some embodiments, the silicon or aluminum abrasive may have a pH level less than 7 for W metal in conductive regions **136** and **140** or may have a pH level greater than 7 for cobalt (Co) or copper (Cu) metals in conductive regions **136** and **140**.

(92) The above embodiments describe structures and methods for simultaneously fabricating semiconductor devices having different fin structures on a same substrate. Such embodiments provide methods of fabricating finFETs, having different fin heights and fin-to-fin pitch with respect to each other, using shared processing steps to simultaneously form one or more components (e.g., STI regions, polysilicon structure, gate structure) of the finFETs. The simultaneous fabrication of finFETs having different configurations of fin structures on the same substrate helps to achieve simpler and more cost-effective fabrication process than the other methods used to fabricate fin structures of different configurations.

(93) In some embodiments, a method of forming first and second finFETs on a substrate includes forming first and second fin structures of the first and second finFETs, respectively, on the substrate. The first and second fin structures have respective first and second vertical dimensions that are about equal to each other. The method further includes modifying the first fin structure such that the first vertical dimension of the first fin structure is smaller than the second vertical dimension of the second fin structure and depositing a dielectric layer on the modified first fin structure and the second fin structure. The method further includes forming a polysilicon structure on the dielectric layer and selectively forming a spacer on a sidewall of the polysilicon structure.

(94) In some embodiments, a method of forming first and second finFETs on a substrate includes forming first and second pair of fin structures of the first and second finFETs, respectively, on the substrate, where a fin-to-fin pitch of the first pair of fin structures is smaller than a fin-to-fin pitch of the second pair of fin structures. The method further includes modifying the first pair of fin structures such that a first vertical dimension of the first pair of fin structures is smaller than a

second vertical dimension of the second pair of fin structures and forming a polysilicon structure over the modified first pair of fin structures and the second pair of fin structures. The method further includes selectively forming a spacer on a sidewall of the polysilicon structure and forming a dielectric layer under the polysilicon structure and the spacer.

(95) In some embodiments, a semiconductor device includes first and second finFETs on a substrate. The first finFET includes a first fin structure having a first vertical dimension and a first epitaxial region on the first fin structure. The second finFET includes a second fin structure having a second vertical dimension that is greater than the first vertical dimension and a second epitaxial region on the second fin structure. The semiconductor device further includes a gate structure over the first and second fin structures, a spacer on the sidewalls of the gate structure, and a dielectric layer under the gate structure and the spacer.

(96) In some embodiments, a method of forming first and second finFETs on a substrate includes forming first and second fin structures of the first and second finFETs, respectively, on the substrate. The first and second fin structures have respective first and second vertical dimensions that are equal to each other. The method further includes modifying the first fin structure such that the first vertical dimension of the first fin structure is smaller than the second vertical dimension of the second fin structure and forming a polysilicon structure over the modified first fin structure and the second fin structure. The method further includes selectively forming a spacer on a sidewall of the polysilicon structure, recessing the modified first fin structure and the second fin structure, and forming epitaxial source/drain regions on the recessed modified first fin structure and the recessed second fin structure.

(97) In some embodiments, a method of forming first and second finFETs on a substrate includes forming first and second pair of fin structures of the first and second finFETs, respectively, on the substrate, where a fin-to-fin pitch of the first pair of fin structures is smaller than a fin-to-fin pitch of the second pair of fin structures. The method further includes modifying the first pair of fin structures such that a first vertical dimension of the first pair of fin structures is smaller than a second vertical dimension of the second pair of fin structures, recessing the modified first fin structure and the second fin structure, and forming a merged epitaxial source/drain region on the recessed modified first fin structure and an epitaxial source/drain region on the recessed second fin structure.

(98) In some embodiments, a semiconductor device includes first and second finFETs on a substrate. The first finFET includes a pair of fin structures having a first vertical dimension and a merged source/drain epitaxial region on the pair of fin structures. The second finFET a second vertical dimension that is greater than the first vertical dimension and a source/drain epitaxial region on the second fin structure. The semiconductor device further includes a gate structure over the first and second fin structures, a spacer on the sidewalls of the gate structure, and a dielectric layer under the gate structure and the spacer.

(99) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate; first and second fin structures having a first vertical dimension disposed on the substrate; a third fin structure having a second vertical

dimension that is greater than the first vertical dimension; first, second, and third source/drain (S/D) regions disposed on the first, second, and third fin structures, respectively; a first silicide layer disposed on and between the first and second S/D regions; a second silicide layer disposed on the third S/D region; a dielectric layer comprising first and second dielectric portions, wherein the first dielectric portion is disposed along sidewalls of the first and second S/D regions that face each other, wherein the second dielectric portion is disposed along sidewalls of the third S/D region, wherein an interface between the first dielectric portion and the first silicide layer is disposed below top surfaces of the first and second fin structures, and wherein an interface between the second dielectric portion and the second silicide layer is disposed below a top surface of the third fin structure; and a gate structure disposed on the first, second, and third fin structures.

2. The semiconductor device of claim 1, wherein a distance between the first and second fin structures is smaller than a distance between the second and third fin structures.
3. The semiconductor device of claim 1, further comprising an isolation region disposed on the substrate, wherein a top surface of the isolation region is below the top surfaces of the first and second fin structures.
4. The semiconductor device of claim 1, wherein the first S/D region surrounds sidewalls of the first fin structure.
5. The semiconductor device of claim 1, wherein the first silicide layer is in contact with the first and second fin structures.
6. The semiconductor device of claim 1, wherein the second silicide layer is disposed along a top surface and sidewalls of the third S/D region.
7. The semiconductor device of claim 1, wherein the first silicide layer is in contact with the second dielectric portion.
8. The semiconductor device of claim 1, wherein an interface between the first silicide layer and the first dielectric portion is disposed above the top surfaces of the first and second fin structures.
9. The semiconductor device of claim 1, wherein a cross-sectional profile of the first silicide layer is tapered along a first direction, and wherein a cross-sectional profile of the second silicide layer is tapered along a second direction that is opposite to the first direction.
10. The semiconductor device of claim 1, further comprising a conductive layer disposed on the first silicide layer, wherein a bottom surface of the conductive layer is disposed below the top surfaces of the first and second fin structures.
11. A semiconductor device, comprising: a substrate; first, second, and third fin structures disposed on the substrate; first, second, and third source/drain (S/D) regions disposed on the first, second, and third fin structures, respectively; a first metal-based structure disposed on and between the first and second S/D regions; a second metal-based structure disposed on the third S/D region; wherein a cross-sectional profile of the first metal-based structure is tapered along a first direction, and wherein a cross-sectional profile of the second metal-based structure is tapered along a second direction that is opposite to the first direction; a first dielectric layer disposed between the first and second metal-based structures and between the second and third S/D regions; an etch stop layer disposed between the first and second S/D regions, wherein a top surface of the etch stop layer is in contact with a bottom surface of a tapered portion of the first metal-based structure; and a gate structure disposed on the first, second, and third fin structures.
12. The semiconductor device of claim 11, wherein the second metal-based structure is disposed along a top surface and sidewalls of the third S/D region.
13. The semiconductor device of claim 11, wherein the first metal-based structure is in contact with the first and second fin structures.
14. The semiconductor device of claim 11, further comprising: an isolation region disposed between the first and second fin structures; and a second dielectric layer disposed between the first metal-based structure and the isolation region.
15. The semiconductor device of claim 11, further comprising an isolation region disposed between

the first and second S/D regions.

16. The semiconductor device of claim 11, further comprising: an isolation region surrounding a portion of the third S/D region; and a second dielectric layer disposed between a bottom surface of the second metal-based structure and a top surface of the isolation region.

17. A semiconductor device, comprising: a substrate; first, second, and third fin structures disposed on the substrate, wherein a distance between the first and second fin structures is smaller than a distance between the second and third fin structures; first, second, and third source/drain (S/D) regions disposed on the first, second, and third fin structures, respectively; a first conductive structure with tapered sidewalls disposed on the first and second S/D region; a second conductive structure with curved sidewalls disposed on the third S/D region; a first dielectric layer disposed along sidewalls of the second and third S/D regions that face each other; a second dielectric layer disposed on the first dielectric layer and sidewalls of the first and second conductive structures; and a gate structure disposed on the first, second, and third fin structures.

18. The semiconductor device of claim 17, wherein the second conductive structure is disposed along a top surface and sidewalls of the third S/D region.

19. The semiconductor device of claim 17, wherein a cross-sectional profile of the first conductive structure is tapered along a first direction, and wherein a cross-sectional profile of the second conductive structure is tapered along a second direction that is opposite to the first direction.

20. The semiconductor device of claim 17, further comprising: an isolation region surrounding a portion of the third S/D region; and a third dielectric layer disposed between a bottom surface of the second conductive structure and a top surface of the isolation region.
